# Governing Simulator Discrepancy via Deterministic Pairing and Horizon-Validated Residual Dynamics

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**Abstract**

Residual learning across simulators is ill-posed if trajectory pairing, state semantics, and rollout timing are not aligned. In robotics pipelines, this creates a silent failure mode: models can look correct on one-step logs but fail dynamically under contact-rich execution. We present a governance-oriented sim-to-sim pipeline for PyBullet–MuJoCo Franka pushing that enforces deterministic pairing at episode/timestep granularity, applies a hard alignment gate before training, and evaluates residual correction under long-horizon stress. The correction is formulated as $\hat{S}_{t+1} = \Pi(S^f_{t+1} + \Delta(S_t, A_t))$, where $\Pi$ maintains valid angles, quaternions, and bounded velocities. We report physical-state and contact-proxy metrics separately to avoid conflating solver-dependent bookkeeping with physically meaningful mismatch. On the latest passing hard-mode run, physical one-step p95 decreases from 0.1196 to 0.0207, and corrected rollout p95 remains stable at horizons 50/200/500 (0.0173/0.0174/0.0222) with zero NaN/Inf/divergence. Contact-regime physical p95 remains low (0.0075 contact, 0.0131 near-contact, 0.0082 force-top-5%). These results support deterministic pairing + alignment gating as practical controls for stable sim-to-sim residual correction under contact.

## 1 Introduction

Robotics teams routinely combine simulators to balance throughput, contact fidelity, and engineering convenience. The practical risk is not only model bias, but governance failure: simulator disagreement can enter training silently when reset conventions, frame semantics, or temporal indexing are not verified. As a result, residual models may appear accurate per-step while failing when executed over horizon.

This work treats cross-simulator correction as a systems validation problem. Rather than assuming paired data is valid, we enforce deterministic plan replay and explicit sample correspondence, then block training with a hard gate when alignment checks fail. We further require horizon-level validation in contact-rich regimes, because one-step loss alone is not a sufficient criterion for dynamic correctness.

Contact interactions are the stress case where hidden mismatch is most likely to surface. Near-contact and high-force phases amplify small semantic differences in collision handling, velocity integration, and state extraction. These effects may be negligible in free-space but accumulate under rollout, making contact slices and long-horizon checks central to trustworthy evaluation.

Figure 1 summarizes the governance pipeline used in this paper.

Our contributions are:

1. A deterministic paired-rollout protocol with strict episode/time index matching.
2. A hard alignment gate that blocks training when reset, timing, or frame consistency fails.

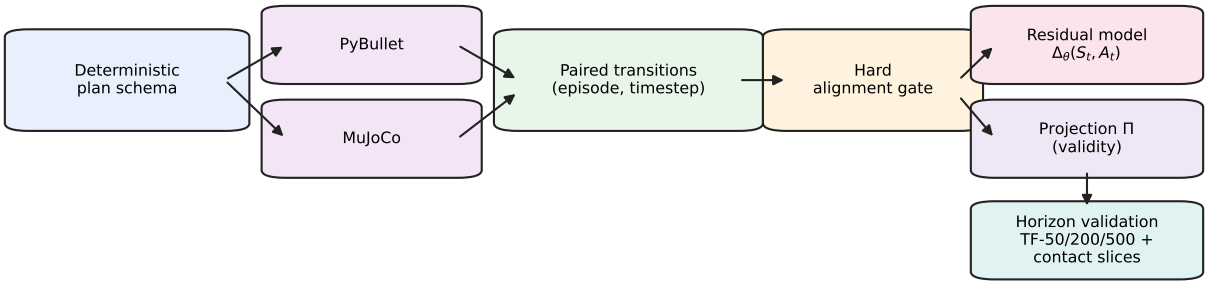


Figure 1: Governance pipeline for sim-to-sim residual correction: deterministic plan replay, paired transitions, hard alignment gate, projected correction, and horizon/contact validation.

3. A residual correction model evaluated under one-step, 50/200/500-step rollouts, and contact-focused slices.

4. A reporting protocol that separates physical-state error from solver-dependent contact-proxy error.

## 2 Related Work

Prior sim-to-real and sim-to-sim work uses domain randomization, system identification, and residual learning [5, 4, 2, 3]. Recent theoretical work on geometry, flatness, and information concentration under structured designs also informs how we think about stability and constrained correction [1]. These approaches are powerful but often under-specify cross-simulator *validation semantics*: paired-index integrity, one-step vs rollout mismatch, and contact-regime stability are frequently conflated.

Our focus is orthogonal: we provide a deterministic measurement-and-gating protocol around residual correction, then evaluate under explicit long-horizon and regime-sliced stress conditions.

## 3 Problem Formulation

Let $S_t \in \mathbb{R}^n$ and $A_t \in \mathbb{R}^m$ denote state and action. We use two simulators: fast simulator $\mathcal{E}_f$ and reference simulator $\mathcal{E}_r$. For paired transitions,

$$(S_t, A_t, S^f_{t+1}, S^r_{t+1}). \tag{1}$$

We learn a residual model

$$\Delta_\theta(S_t, A_t) \approx S^r_{t+1} - S^f_{t+1}, \tag{2}$$

and corrected next state

$$\hat{S}_{t+1} = \Pi\left(S^f_{t+1} + \Delta_\theta(S_t, A_t)\right), \tag{3}$$

where $\Pi$ is a projection operator that enforces state validity.

**Projection $\Pi$.** We apply quaternion renormalization, joint-angle wrapping to $[-\pi, \pi]$, bounded velocity clipping, and bounded contact metadata clipping. This is used both in training and rollout evaluation.

# 4 Deterministic Pairing and Alignment Gate

## 4.1 Deterministic paired data

Both simulators execute the same deterministic plan schema with shared episode IDs and per-step timestamps. Pairing validity requires: identical sample count, episode monotonicity, and unit timestep increments.

## 4.2 Hard alignment gate

Training is blocked unless alignment checks pass:

- reset consistency under shared plan,
- effective step-time consistency,
- frame-consistent state extraction,
- strict early-step drift tolerance.

# 5 Residual Models

We evaluate a conditioned mixture-of-experts residual model with regime-aware gating. Inputs are $(S_t^f, A_t)$ plus family and parameter conditioning metadata. Training uses robust Huber loss on corrected next state, guarded gradients, and projection-consistent updates.

# 6 Evaluation Protocol

## 6.1 Metric spaces

We report in original units with disjoint masks:

- **Physical-state dims**: robot, end-effector, object pose/velocity.
- **Contact-proxy dims**: contact flag, contact force proxy, min distance.

Pass/fail uses physical-state metrics only; contact-proxy metrics are separate diagnostics. If contact proxies are non-finite or contact-flag agreement collapses, verdict fails.

## 6.2 Stress protocol

- One-step physical error (mean/p95/max).
- Teacher-forced rollout physical error at horizons 50/200/500.
- Free-running stability rollout (NaN/Inf/divergence checks).
- Regime slices: contact, near-contact, force-top-5%, action quantiles, temporal slices.

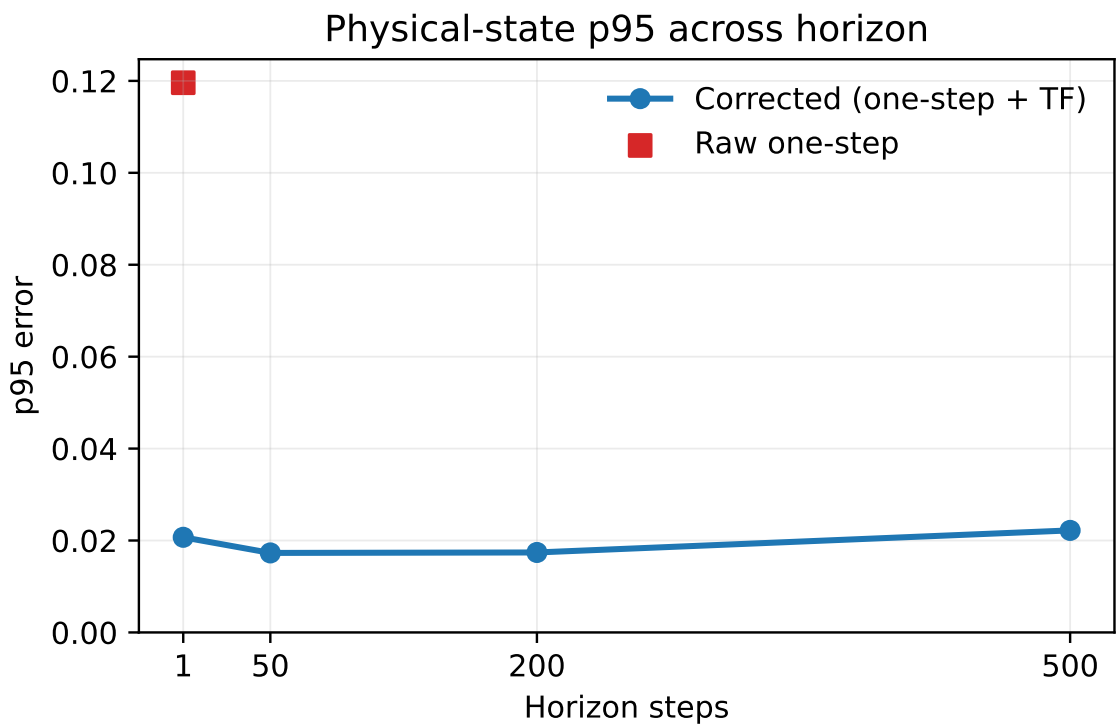


Figure 2: Physical-state p95 vs horizon. Corrected error remains low from one-step to 500-step teacher-forced rollout.

- Distribution shift: seed holdout (D1) and family holdout (D2).

Figure 2 visualizes horizon-wise p95 behavior, and Figure 3 summarizes contact-slice p95 values.

## 7 Results

Table 1 reports the final hard-mode run (PASS) from `reports/stress_20260203-173231`.

**Contact-regime physical slices.**

- contact1 p95: 0.0075
- near-contact p95: 0.0131
- force-top-5% p95: 0.0082

All required hard-mode thresholds pass.

**Contact-proxy diagnostics (separate).** Contact-flag accuracy is 0.9911; proxy p95 values are bounded (force: 0.1236, min-dist: 0.0046).

**Distribution shift.** Both D1 seed holdout and D2 family holdout pass in hard mode.

**Stability.** Free-running stability rollouts at 50/200/500 steps show zero NaN/Inf/divergence.

Table 1: Main physical-state results (original units).

| Metric | Mean | p95 | Max |
|---|---|---|---|
| Raw one-step (PHYS) | 0.0941 | 0.1196 | 0.4627 |
| Corrected one-step (PHYS) | 0.0106 | 0.0207 | 0.1006 |
| Rollout TF-50 (PHYS) | 0.0100 | 0.0173 | 0.0844 |
| Rollout TF-200 (PHYS) | 0.0099 | 0.0174 | 0.0821 |
| Rollout TF-500 (PHYS) | 0.0110 | 0.0222 | 0.0872 |

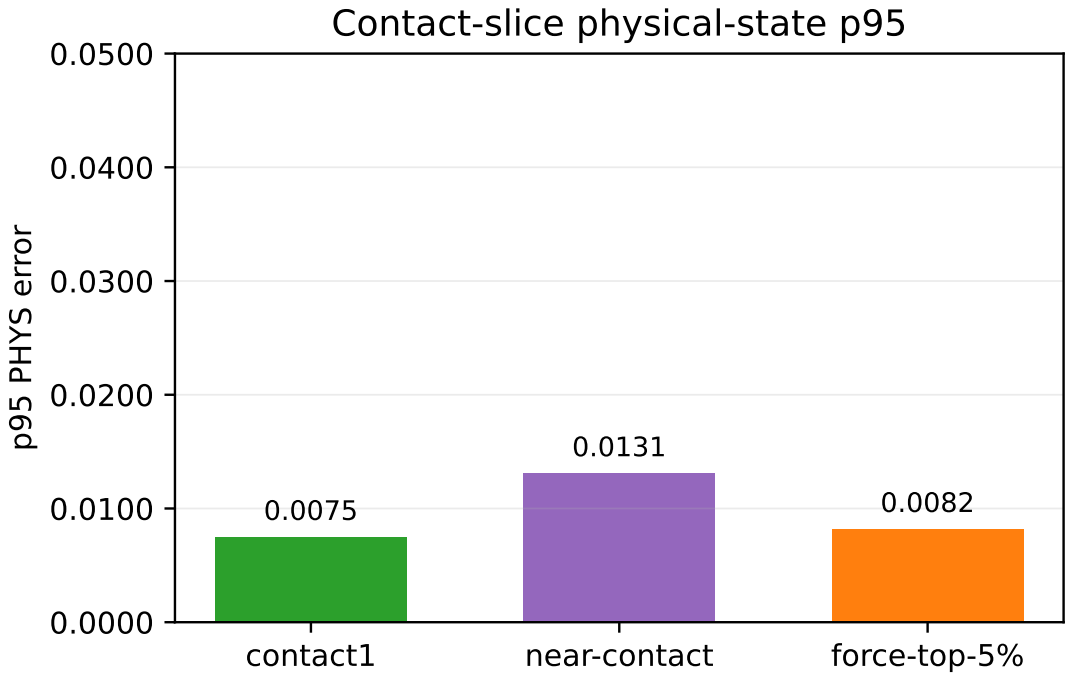


Figure 3: Contact-slice physical-state p95 values remain low across contact1, near-contact, and force-top-5% regimes.

## 8 Discussion

The dominant practical insight is semantic consistency: projection-corrected residual models can remain stable only when state semantics, pairing, and evaluation loops are explicitly aligned. Without this governance layer, one-step gains can coexist with rollout failure.

## 9 Limitations

This work is sim-to-sim only, on one robot/task family (Franka push), with deterministic plan generation. We do not claim sim-to-real transfer.

## 10 Conclusion

Simulator mismatch can be governed through deterministic pairing, alignment gating, projection-consistent residual correction, and horizon-validated stress testing. Under this protocol, cross-simulator correction remains accurate and stable in contact-rich long-horizon settings.

## References


[1] O. Anthony. A universal curvature–information principle: Flatness and $d^{-1}$ concentration under 2-designs, 2025.

[2] Stephen James and et al. Sim-to-real via sim-to-sim: Data-efficient robotic grasping via randomized-to-canonical adaptation networks. *CVPR*, 2019.

[3] Fabio Muratore and et al. Data-efficient domain randomization with bayesian optimization. *IEEE Robotics and Automation Letters*, 2021.

[4] Xue Bin Peng, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. Sim-to-real transfer of robotic control with dynamics randomization. *ICRA*, 2018.

[5] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *IROS*, 2017.


## A Reproducibility Artifacts

Primary passing run: `reports/stress_20260203-173231`.
Model: `reports/delta_train_hardgen_fix_20260203-135301/model.pt`.
Datasets: `data_pybullet/paired_bullet_hardgen_seeds0-5.npz`, `data_mujoco/paired_mujoco_hardgen_seeds`